ASSOCIATION RULES: MARKET BASKET ANALYSIS

Outline

- Discuss applications of association rules
- Conduct market basket analysis
- Describe mathematical criteria for evaluating rules
- Explain the importance of domain expertise for interpreting association rules

Association Rules: Market Basket Analysis

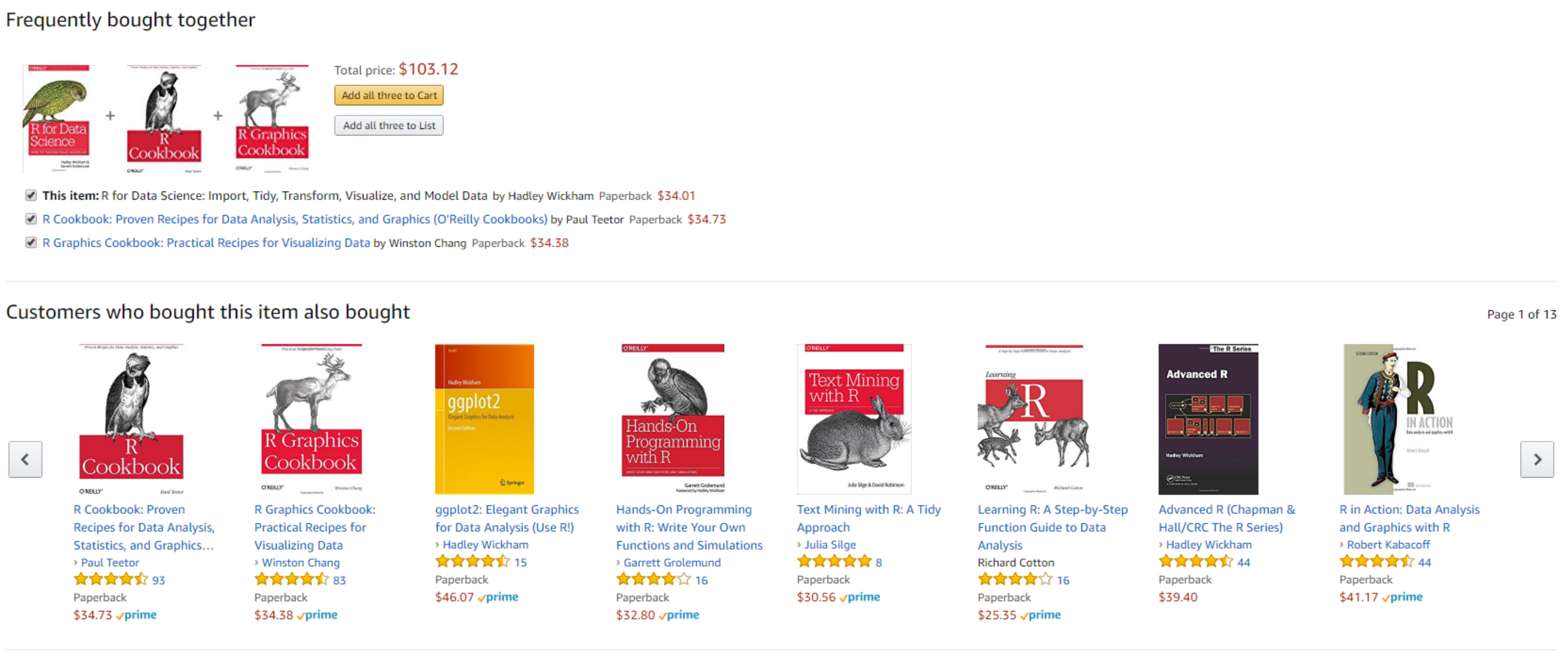
- Market basket analysis is …
- …. a data mining technique that has the purpose of finding the optimal combination of products or services and allows marketers to exploit this knowledge to provide recommendations, optimize product placement, or develop marketing programs that take advantage of cross-selling.
- … a technique to see which items go together
- … widely used by retailers to identify items that are purchased in the same shopping trip.
- … not limited to retailers. May also be used by insurance companies, banks, etc.

Analytical Technique

- Unsupervised Learning
- Based on numeric thresholds, not statistics
- Involves working with large sparse matrices

ILLUSTRATIONS

One of the most quoted illustrations of market basket analysis is a supermarket chain that found, “*male customers that bought diapers often bought beer as well*”, so “*they put the diapers close to beer coolers, and their sales increased dramatically*” (Wikipedia).



Source: Amazon.com webpage for Vishal Lala

Source: Found on http://www.theexaminingroom.com/ in Sept, 2012

SMALL DATA

Data

beer	diapers	bread	
diapers	eggs		
diapers	beer		
beer	diapers	eggs	
beer	diapers		
diapers	milk		
milk	bread		
diapers	beer	milk	bread
beer	diapers	milk	

Data

beer	diapers	bread	
diapers	eggs		
diapers	beer		
beer	diapers	eggs	
beer	diapers		
diapers	milk		
milk	bread		
diapers	beer	milk	bread
beer	diapers	milk	

transactions in sparse format with
9 transactions (rows) and
5 items (columns)

most frequent items:

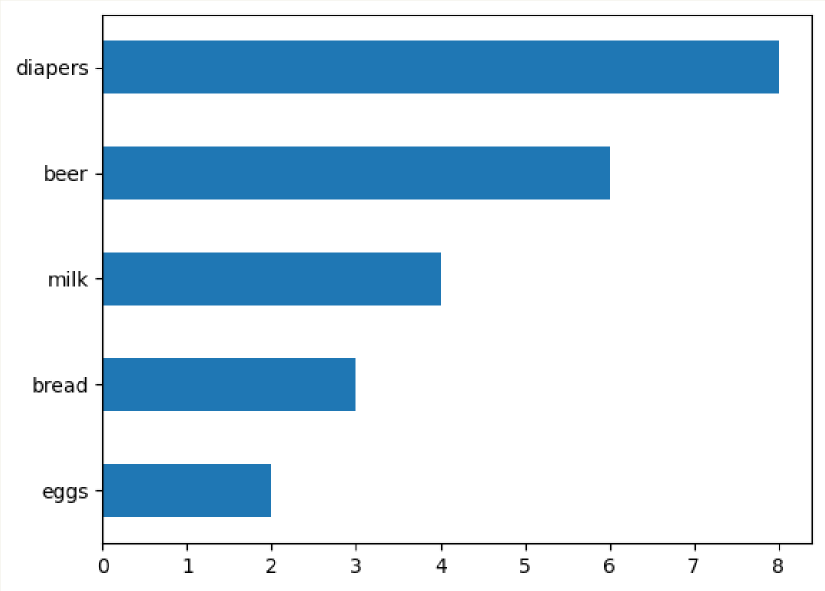
diapers beer milk bread eggs (Other)

8 6 4 3 2
0

Data

beer	diapers	bread	
diapers	eggs		
diapers	beer		
beer	diapers	eggs	
beer	diapers		
diapers	milk		
milk	bread		
diapers	beer	milk	bread
beer	diapers	milk	

Item Frequency Plot



Data

beer	diapers	bread	
diapers	eggs		
diapers	beer		
beer	diapers	eggs	
beer	diapers		
diapers	milk		
milk	bread		
diapers	beer	milk	bread
beer	diapers	milk	

Many Rules

For k items, number of rules is of the order 2^k

We are interested in selecting rules that indicate a strong association (i.e., high lift) and are not rare (i.e., high support)

	antecedents	consequents
0	(beer)	(bread)
1	(bread)	(beer)
2	(beer)	(diapers)
3	(diapers)	(beer)
4	(milk)	(bread)
5	(bread)	(milk)
6	(diapers)	(eggs)
7	(eggs)	(diapers)
8	(beer, diapers)	(bread)
9	(beer, bread)	(diapers)
10	(diapers, bread)	(beer)
11	(beer)	(diapers, bread)
12	(diapers)	(beer, bread)
13	(bread)	(beer, diapers)
14	(beer, milk)	(bread)

Measuring Affinity

- Rule: A → B
- Support: p(A&B)
  - *Prevalence of an item set*
- Confidence: p(B|A)
  - *Predictability of an association rule*
- Lift: p(B|A)/p(B) = p(A&B)/(p(A)*p(B))
  - *Strength of association. Lift of 1 indicates independence, i.e., no association.*

Measuring Affinity

Rule	Support	Confidence
$A \Rightarrow D$	2/5	2/3
$C \Rightarrow A$	2/5	2/4
$A \Rightarrow C$	2/5	2/3
$B \& C \Rightarrow D$	1/5	1/3

Source: Found on https://themainstreamseer.blogspot.com/2012/09/numeric-measures-for-association-rules.html on Sept 2012

Data

beer	diapers	bread	
diapers	eggs		
diapers	beer		
beer	diapers	eggs	
beer	diapers		
diapers	milk		
milk	bread		
diapers	beer	milk	bread
beer	diapers	milk	

Consider the Rule: Diapers → Beer

Rule Frequency: How often do diapers and beer co-occur? 6

Support, p(Beer & Diapers): What proportion of transactions contain both diapers and beer? 6/9

Confidence, p(Beer|Diapers): Given that diapers occur in a transaction, what is the chance the transaction contains beer? 6/8

Lift = p(Beer|Diapers)/p(Beer)

= p(Beer & Diapers)/p(Beer)p(Diapers)

= (6/9)/((6/9)*(8/9))

AMAZON DATA

Medium Data

- This data is derived from a dataset of Amazon Product Reviews.
- The dataset describes 25000 orders on more than 100 unique products spread across seven categories.
- Each row in the data is a transaction and includes order id and products purchased.

	Product	Count
0	Fire TV Stick 4K Max	953
1	KitchenAid Stand Mixer	934
2	Maybelline Mascara	909
3	Ring Video Doorbell Pro	902
4	Adidas Ultraboost 22	899
5	LEGO Star Wars Millennium Falcon	893
6	Nike Soccer Ball	893
7	Fitbit Charge 6	888
8	Logitech MX Master 3S Mouse	882
9	Sony WH-1000XM5 Headphones	881

More on Market Basket Analysis

- Market Basket Analysis generates a large number of rules. It is for the domain expert to identify the rules that are relevant and meet association thresholds
- Rules can be reduced by
  - *grouping items using domain knowledge.*
  - *text analysis. E.g., combining all ring doorbells.*
  - *clustering.*

Summary

- In this module we,
  - *discussed applications of association rules*
  - *Conducted market basket analysis*
  - *Described mathematical criteria for evaluating rules*
  - *Explained the importance of domain expertise for interpreting association rules*